A Perturbed Constant-Momentum Method Escapes Strict Saddle Points

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Abstract

We study a perturbed momentum method with a constant momentum parameter, a fixed step size, and a restart safeguard, and prove that it converges to an approximate second-order stationary point of a smooth nonconvex objective. Writing L_1 and L_2 for the gradient- and Hessian-Lipschitz constants and $(\varepsilon_g, \varepsilon_H)$ for the first- and second-order tolerances, the method finds a point with $\|\nabla f(x)\| \le \varepsilon_g$ and $\lambda_{\min}(\nabla^2 f(x)) \ge -\varepsilon_H$ in

$$O\left(\frac{L_1\,\Delta_f}{\varepsilon_g^2}\right) + \widetilde{O}\left(\frac{L_1^3 L_2^2\,\Delta_f}{\varepsilon_H^4}\right)$$

iterations with high probability, where Δ_f bounds the initial suboptimality and $\widetilde{O}$ hides polylogarithmic factors in the dimension and the failure probability. Under the standard scaling $\varepsilon_H = \sqrt{L_2 \varepsilon_g}$ this matches the $\widetilde{O}(\varepsilon_g^{-2})$ rate of perturbed gradient descent. Two ingredients drive the analysis. First, a restart test on the momentum direction forces every search direction to be a descent direction with a quantitative angle bound, which yields a per-step decrease and an improve-or-localize estimate. Second, around a strict saddle we linearise the *difference* of two coupled runs that share their perturbation but are offset along the escape direction; the difference obeys a non-symmetric matrix recursion whose dominant eigenvalue we identify explicitly, and a thin-slab volume argument shows that the set of perturbations from which the method fails to escape is negligible.

1 Introduction

Nonconvex optimisation problems are pervasive in modern machine learning, and although locating a global minimiser is intractable in the worst case, a more modest target has proved both useful and attainable: convergence to a stationary point. A point with small gradient may, however, be a saddle point or even a local maximum, and the loss surfaces arising in applications are often densely populated by such points. It is therefore natural to ask for convergence to a *second-order* stationary point, where in addition to a small gradient the Hessian is approximately positive semidefinite, ruling out the strict saddle points at which $\lambda_{\min}(\nabla^2 f) < 0$.

For gradient descent (GD), the analysis of Jin, Ge, Netrapalli, Kakade and Jordan [3] shows that a single random perturbation, injected whenever the gradient is small, suffices to escape strict saddles, and that the perturbed method reaches an approximate second-order stationary point in $\widetilde{O}(\varepsilon^{-2})$ iterations—the same rate, up to logarithmic factors, at which GD reaches a first-order stationary point. The companion work of Jin, Netrapalli and Jordan [4] extends the perturbation idea to Nesterov's accelerated gradient method and isolates two reusable tools: an *improve-or-localize* principle that confines slowly-progressing iterates to a small ball, and a *coupling* argument that bounds the volume of the "stuck" region by tracking two trajectories separated along the escape direction.

The present paper carries this programme to a method with *constant* momentum. We study the iteration $x_{k+1} = x_k + \alpha\, d_k$, $d_{k+1} = -g_{k+1} + \beta\, d_k$ with a fixed coefficient β—equivalently, the heavy-ball recursion $x_{k+1} = x_k - \alpha\nabla f(x_k) + \beta(x_k - x_{k-1})$ [5]—together with a restart safeguard that keeps the momentum direction aligned with the negative gradient, and a perturbation that is injected when the gradient is small. Two features distinguish the analysis from the gradient-descent case. The restart keeps every step a descent step despite the momentum; and, unlike gradient descent, whose linearisation is symmetric, the constant-momentum iteration linearises to a *non-symmetric* propagator that does not diagonalise in the eigenbasis of the Hessian, which we control by an explicit spectral analysis of the two-trajectory difference. Keeping β constant is a deliberate restriction: it isolates the geometric core of the argument and sets aside the gradient-dependent coefficient of genuine nonlinear conjugate gradient, whose denominator $\|\nabla f\|^2$ vanishes at stationary points (see Section 9).

Our analysis rests on two ideas. The first is local: a restart test of the form $\|g_k + d_k\| \le \nu\|g_k\|$ makes every search direction a descent direction with a quantitative angle bound, valid both on momentum and on restart steps; together with the fixed step size $\alpha = 1/L_1$ this gives a per-step decrease proportional to $\|g_k\|^2$ and an improve-or-localize estimate. The second is the coupling: near a strict saddle we follow two runs that share the perturbation but are offset along the smallest curvature direction, linearise their difference, and show that the non-symmetric propagator has a dominant real eigenvalue $\mu_+ \ge 1 + \alpha\varepsilon_H > 1$ that drives the two runs apart, while a thin-slab volume argument bounds the set of perturbations from which the method stays stuck. Section 2 fixes the setting, the method and the main theorem; Section 3 establishes the descent and localization estimates; Section 4 sets up the linearized two-trajectory recursion; Section 5 analyses the propagator and identifies the escape eigenvalue; Section 6 shows the geometric growth survives the nonlinearity; Section 7 turns this into a high-probability escape guarantee; and Section 8 assembles the iteration complexity.

We emphasise that the method is not accelerated: the rate we obtain matches perturbed gradient descent [3], not the $\widetilde{O}(\varepsilon^{-7/4})$ of perturbed AGD [4]. Our contribution is to show that the perturbation-and-coupling machinery extends to a constant-momentum iteration, whose non-symmetric propagator we analyse explicitly, once the momentum direction is safeguarded by restarts.

Related work.

Notation. Throughout, $\|\cdot\|$ is the Euclidean norm for vectors and the spectral norm for matrices, and $\lambda_{\min}(\cdot)$ is the smallest eigenvalue of a symmetric matrix. We write $g_k := \nabla f(x_k)$, $\nabla^2 f$ for the Hessian, and $B_0(r)$ for the Euclidean ball of radius r centred at the origin. We use $O(\cdot)$ and $\Theta(\cdot)$ in the usual way and $\widetilde{O}(\cdot)$ to hide polylogarithmic factors in all problem parameters. The symbol $\tilde{x}$ denotes the snapshot at which a perturbation is added (the candidate saddle), and $x^\star$ a global minimiser. Constants introduced along the way are collected in Table 1.

2 Setting, Algorithm and Main Result

We consider the unconstrained problem $\min_{x\in\mathbb{R}^d} f(x)$ under the following standing assumptions.

Assumption 2.1. $f\colon \mathbb{R}^d \to \mathbb{R}$ is twice continuously differentiable and L_1-gradient-Lipschitz: $\|\nabla f(x) - \nabla f(y)\| \le L_1\|x - y\|$ for all x, y.

Assumption 2.2. f is L_2-Hessian-Lipschitz: $\|\nabla^2 f(x) - \nabla^2 f(y)\| \le L_2\|x - y\|$ for all x, y.

Assumption 2.3. f is bounded below, and $\Delta_f := f(x_0) - \inf_x f(x) < \infty$.

Definition 2.1 (Approximate second-order stationary point)**.** Given tolerances $\varepsilon_g, \varepsilon_H > 0$, a point x is an $(\varepsilon_g, \varepsilon_H)$-second-order stationary point (2OSP) of f if

$$\|\nabla f(x)\| \le \varepsilon_g \qquad \text{and} \qquad \lambda_{\min}\big(\nabla^2 f(x)\big) \ge -\varepsilon_H.$$

Algorithm 1 PR-CMOM: Perturbed Restarted Constant Momentum PR-CMOM(x_0, L_1, L_2, $\varepsilon_g, \varepsilon_H$, β, ν, δ_{tot}, Δ_f)

Derived constants (resolve the constraint system of Section 8):
  $\alpha \leftarrow 1/L_1$; $\Lambda \leftarrow \log\frac{d\,L_1 L_2 \Delta_f}{\varepsilon_g\,\varepsilon_H\,\delta_{\text{tot}}}$; $c(\nu) \leftarrow (1-\nu) - \frac{(1+\nu)^2}{2}$
  $\mu_+ \leftarrow$ larger root of $\mu^2 - (1+\beta+\alpha\varepsilon_H)\mu + \beta$; resolve (T, S, F, r) and per-episode $\delta = \delta_{\text{tot}}/\bar{K}$
$d_0 \leftarrow -\nabla f(x_0)$; $t_{\text{noise}} \leftarrow -T-1$
for $t = 0, 1, 2, \ldots$ **do**
  if $\|\nabla f(x_t)\| \le \varepsilon_g$ **and** $t - t_{\text{noise}} > T$ **then** ▷ gradient small: test for negative curvature
    $\tilde{x} \leftarrow x_t$; $t_{\text{noise}} \leftarrow t$
    $x_t \leftarrow \tilde{x} + \xi_t$, $\xi_t \sim \text{Unif}\big(\mathcal{B}^d(0, r)\big)$ ▷ perturb
    $d_t \leftarrow -\nabla f(x_t)$ ▷ reset momentum at the snapshot
  end if
  if $t - t_{\text{noise}} = T$ **and** $f(x_t) - f(\tilde{x}) > -F$ **then**
    return $\tilde{x}$ ▷ no decrease over the window: $\tilde{x}$ is an $(\varepsilon_g, \varepsilon_H)$-2OSP
  end if
  $x_{t+1} \leftarrow x_t + \alpha\, d_t$ ▷ fixed step $\alpha = 1/L_1$
  $g_{t+1} \leftarrow \nabla f(x_{t+1})$ ▷ **constant** momentum
  $\hat{d}_{t+1} \leftarrow -g_{t+1} + \beta\, d_t$ ▷ candidate conjugate direction
  if $\|\hat{d}_{t+1} + g_{t+1}\| \le \nu\, \|g_{t+1}\|$ **then**
    $d_{t+1} \leftarrow \hat{d}_{t+1}$ ▷ keep momentum (restart test passes)
  else
    $d_{t+1} \leftarrow -g_{t+1}$ ▷ restart: drop momentum
  end if
end for

The single-tolerance formulation of [?, 4] sets $\varepsilon_H = \sqrt{L_2\,\varepsilon}$ and $\varepsilon_g = \varepsilon$; we keep the two tolerances separate to follow the formulation of [1].

The method. The method is the constant-momentum iteration

$$x_{k+1} = x_k + \alpha\, d_k, \qquad d_k = -g_k + \beta\, d_{k-1},$$

with a *fixed* momentum coefficient $\beta \in (0, 1]$ and step size $\alpha = 1/L_1$, together with two safeguards. Eliminating the search direction gives the equivalent heavy-ball form [5]

$$x_{k+1} = x_k - \alpha\, \nabla f(x_k) + \beta\,(x_k - x_{k-1}).$$

- *Restart test.* The momentum direction is accepted only if it is sufficiently aligned with the negative gradient,
$$\|g_k + d_k\| \le \nu\, \|g_k\|, \tag{1}$$
with slack $\nu \in (0, \sqrt{5} - 2)$; otherwise the method restarts, setting $d_k = -g_k$. We write $\mathcal{N}$ for the steps on which the momentum direction is accepted (momentum steps) and $\mathcal{R}$ for the restart steps; on $\mathcal{R}$ the test (1) holds with $\nu = 0$.

- *Perturbation.* When $\|g_k\| \le \varepsilon_g$ and no perturbation has been added in the last T iterations, the iterate is perturbed, $x_k \leftarrow x_k + \xi$ with $\xi \sim \text{Unif}(B_0(R))$, and the direction is reset.

The step size, restart slack, momentum β, perturbation radius R, window length T and gradient trigger ε_g are fixed in Section 8; their dependence on the problem data is recorded in Table 1.

Our main result is the following; its proof occupies Sections 3 to 7 and is assembled in Section 8.

Theorem 2.1 (Main result)**.** *Let Assumptions 2.1 to 2.3 hold and fix a total failure probability $\delta_{\mathrm{tot}} \in (0,1)$. There is a choice of parameters (Section 8) under which, provided each escape window is a momentum window (Definition 4.1), the method visits an $(\varepsilon_g, \varepsilon_H)$-second-order stationary point with probability at least $1-\delta_{\mathrm{tot}}$ after at most*

$$N \;=\; \underbrace{O\!\left(\frac{L_1\,\Delta_f}{C(\nu)\,\varepsilon_g^2}\right)}_{\text{large-gradient phase}} \;+\; \underbrace{\widetilde{O}\!\left(\frac{L_1^3 L_2^2\,\Delta_f}{\varepsilon_H^4}\right)}_{\text{escape phase}} \;=\; O(\varepsilon_g^{-2}) + \widetilde{O}(\varepsilon_H^{-4})$$

iterations, where $C(\nu) = \frac{1}{2}(1-4\nu-\nu^2) > 0$ and $\widetilde{O}$ hides factors polylogarithmic in d, $1/\delta_{\mathrm{tot}}$ and the problem constants.

Remark 2.1. Under the coupling $\varepsilon_H = \sqrt{L_2\,\varepsilon_g}$ of Definition 2.1, $\varepsilon_H^{-4} = (L_2\varepsilon_g)^{-2}$ and the escape phase costs $\widetilde{O}(\varepsilon_g^{-2})$, so the overall rate is $\widetilde{O}(\varepsilon_g^{-2})$, matching perturbed gradient descent [3, Theorem 7].

Remark 2.2. The escape-phase bound is established on momentum windows (Definition 4.1); whether restarts can be ruled out inside an escape window is discussed in Section 9.

Proof strategy near a saddle. The large-gradient phase is handled by a counting argument (Section 8). The crux is the escape phase. Suppose the method perturbs at a snapshot $\tilde{x}$ with $\lambda_{\min}(\nabla^2 f(\tilde{x})) \le -\varepsilon_H$. We run two coupled copies from $x_0 = \tilde{x} + \xi$ and $x_0' = x_0 + R_0 v_{\min}$, sharing the perturbation but offset by R_0 along the smallest curvature direction $v_{\min}$. *(I)* Their difference obeys a linear recursion whose homogeneous part grows like $\mu_+^k R_0$ with $\mu_+ \ge 1 + \alpha\varepsilon_H > 1$, and the nonlinear residual stays a fixed fraction of it, so the gap grows geometrically. *(II)* A copy that fails to escape within T steps is confined, by improve-or-localize, to a ball of radius S; two confined copies cannot be farther than $2S + R_0$ apart, which caps the separation of any two stuck starts at a threshold $R_0^\star$. *(III)* The stuck set is therefore a slab of width $R_0^\star$ along $v_{\min}$, and *(IV)* a uniform perturbation lands in it with probability at most $(R_0^\star/R)\sqrt{(d+1)/(2\pi)}$, which we drive below δ. Repeating over the at most $\widetilde{O}(\varepsilon_H^{-3})$ saddle episodes and counting the large-gradient steps gives Theorem 2.1.

3 A Descent Direction and Localization

We first turn the restart test into the two estimates that the rest of the analysis rests on: a quantitative descent property of the search direction, and the consequent per-step decrease.

Lemma 3.1 (Direction bounds)**.** *On every iteration, whether a momentum step ($\mathcal{N}$) or a restart step ($\mathcal{R}$),*

$$\|d_k\| \le (1+\nu)\|g_k\| \qquad \text{and} \qquad \langle d_k,\, g_k\rangle \le -(1-\nu)\|g_k\|^2. \tag{2}$$

Proof. The accepted direction satisfies $\|g_k + d_k\| \le \nu\|g_k\|$ (on restart steps with $\nu = 0$, since $d_k = -g_k$). For the norm,

$$\|d_k\| = \|(d_k + g_k) - g_k\| \le \|d_k + g_k\| + \|g_k\| \le (1+\nu)\|g_k\|.$$

For the inner product, write $d_k = (d_k + g_k) - g_k$, so that

$$\langle d_k,\, g_k\rangle = -\|g_k\|^2 + \langle d_k + g_k,\, g_k\rangle \le -\|g_k\|^2 + \|d_k + g_k\|\,\|g_k\| \le (\nu - 1)\|g_k\|^2. \qquad \square$$

Lemma 3.2 (Per-step decrease)**.** *Let Assumption 2.1 hold and let $\alpha = 1/L_1$. For every non-perturbation step,*

$$f(x_{k+1}) \le f(x_k) - \frac{C(\nu)}{L_1}\,\|g_k\|^2, \qquad C(\nu) = (1-\nu) - \frac{(1+\nu)^2}{2} = \frac{1-4\nu-\nu^2}{2}, \tag{3}$$

and $C(\nu) > 0$ if and only if $\nu < \sqrt{5} - 2$.

Proof. By L_1-smoothness and $x_{k+1} = x_k + \alpha d_k$,

$$f(x_{k+1}) \le f(x_k) + \alpha \langle g_k, d_k \rangle + \frac{L_1}{2}\alpha^2 \|d_k\|^2.$$

Insert the two bounds of Lemma 3.1:

$$f(x_{k+1}) \le f(x_k) - \alpha(1-\nu)\|g_k\|^2 + \frac{L_1}{2}\alpha^2(1+\nu)^2\|g_k\|^2 = f(x_k) + \|g_k\|^2\left[-\alpha(1-\nu) + \frac{L_1}{2}\alpha^2(1+\nu)^2\right].$$

With $\alpha = 1/L_1$ the bracket equals $-\frac{1}{L_1}\left[(1-\nu) - \frac{1}{2}(1+\nu)^2\right] = -C(\nu)/L_1$. Finally $C(\nu) = \frac{1}{2}(2-2\nu-(1+\nu)^2) = \frac{1}{2}(1-4\nu-\nu^2)$ is positive exactly when $\nu^2+4\nu-1 < 0$, i.e. $\nu < \sqrt{5}-2$. $\square$

The decrease (3) also controls how far the iterates can travel, which is the improve-or-localize principle: either the objective drops appreciably, or the iterates stay in a small ball.

Lemma 3.3 (Improve or localize)**.** *Under the setting of Lemma 3.2, suppose only momentum and restart steps are taken on $[0, T]$. Then for every $k \le T$,*

$$\|x_k - x_0\| \le (1+\nu)\sqrt{\frac{T\,(f(x_0) - f(x_T))}{L_1\,C(\nu)}}. \tag{4}$$

In particular, if $f(x_0) - f(x_T) \le F$ then $\|x_k - x_0\| \le S$ for all $k \le T$, where

$$S \;:=\; (1+\nu)\sqrt{\frac{TF}{L_1\,C(\nu)}}. \tag{5}$$

Proof. Using $x_{k+1} - x_k = \alpha d_k$, the bound $\|d_k\| \le (1+\nu)\|g_k\|$, and (3) in the form $\|g_k\|^2 \le \frac{L_1}{C(\nu)}(f(x_k) - f(x_{k+1}))$,

$$\|x_{k+1} - x_k\|^2 = \alpha^2\|d_k\|^2 \le \alpha^2(1+\nu)^2\|g_k\|^2 \le \frac{(1+\nu)^2}{L_1 C(\nu)}\big(f(x_k) - f(x_{k+1})\big),$$

since $\alpha^2 = 1/L_1^2$. Summing telescopes:

$$\sum_{j=0}^{k-1}\|x_{j+1} - x_j\|^2 \le \frac{(1+\nu)^2}{L_1 C(\nu)}\big(f(x_0) - f(x_k)\big).$$

By the triangle and Cauchy–Schwarz inequalities,

$$\|x_k - x_0\| \le \sum_{j=0}^{k-1}\|x_{j+1} - x_j\| \le \sqrt{k}\left(\sum_{j=0}^{k-1}\|x_{j+1} - x_j\|^2\right)^{1/2} \le (1+\nu)\sqrt{\frac{k\,(f(x_0) - f(x_k))}{L_1 C(\nu)}},$$

and $k \le T$, $f(x_0) - f(x_k) \le f(x_0) - f(x_T) \le F$ give the claim. $\square$

4 Linearizing Two Coupled Trajectories

We now set up the coupling. Fix a snapshot $\tilde{x}$ at which a perturbation is added, and consider two runs of the method,

$$x_0 = \tilde{x} + \xi, \qquad x_0' = x_0 + R_0\, v_{\min},$$

that share the perturbation ξ but are offset by R_0 along the minimal curvature direction $v_{\min}$ of $H := \nabla^2 f(\tilde{x})$. Write $\Delta x_k = x_k - x_k'$ and $\Delta d_k = d_k - d_k'$, and study

$$w_k := \begin{pmatrix} \Delta x_k \\ \Delta d_{k-1} \end{pmatrix} \in \mathbb{R}^{2d}.$$

The key is to write the gradient as its linear part plus a remainder, $g_k = Hx_k + e_k$ with $e_k := \nabla f(x_k) - Hx_k$, and to control the remainder through the Hessian-Lipschitz constant.

Definition 4.1 (Momentum window)**.** A block of T iterations following a perturbation is a *momentum window* if, for both coupled trajectories, the restart test (1) fails to trigger at every step, so that $d_k = -g_k + \beta\, d_{k-1}$ holds with the same fixed β throughout. Equivalently, the propagator $\mathbf{M}$ below is constant across the window.

The escape phase is analysed on momentum windows; this is the analogue of the "only accelerated steps, no negative-curvature exploitation" regime of [4]. A restart replaces $\mathbf{M}$ by the matrix obtained by setting $\beta = 0$, so the single-matrix growth estimates of Section 5 no longer apply; discharging Definition 4.1—showing restarts are inactive near a saddle—is left to future work (Section 9).

Lemma 4.1 (Difference recursion)**.** *Along a momentum window (Definition 4.1), with $\alpha = 1/L_1$, and the constant momentum β,*

$$w_{k+1} = \mathbf{M}\, w_k + \mathcal{R}_k, \qquad \mathbf{M} = \begin{pmatrix} I - \alpha H & \alpha\beta I \\ -H & \beta I \end{pmatrix}, \qquad \mathcal{R}_k = \begin{pmatrix} -\alpha\, \Delta e_k \\ -\Delta e_k \end{pmatrix}, \tag{6}$$

where $\Delta e_k = \nabla f(x_k) - \nabla f(x_k') - H\Delta x_k$. Moreover, if all iterates of both runs stay within S of their starts and within R of $\tilde{x}$ at the start, then

$$\|\mathcal{R}_k\| \le \sigma_{\mathrm{err}}\, \|w_k\|, \qquad \sigma_{\mathrm{err}} := \sqrt{1+\alpha^2}\; L_2(S+R). \tag{7}$$

Proof. From $x_{k+1} = x_k + \alpha d_k$ and $d_k = -g_k + \beta d_{k-1} = -(Hx_k + e_k) + \beta d_{k-1}$ we obtain, on differencing the two runs,

$$\Delta d_k = -H\Delta x_k - \Delta e_k + \beta\, \Delta d_{k-1}, \qquad \Delta x_{k+1} = \Delta x_k + \alpha\, \Delta d_k.$$

Substituting the first into the second,

$$\Delta x_{k+1} = (I - \alpha H)\Delta x_k + \alpha\beta\, \Delta d_{k-1} - \alpha\, \Delta e_k,$$

which is the top block of (6); the bottom block is the displayed expression for Δd_k. This proves the recursion.

For the residual, set $E(x) := \nabla f(x) - Hx$, whose Jacobian is $J_E(x) = \nabla^2 f(x) - H$, so that by Assumption 2.2, $\|J_E(x)\| \le L_2\|x - \tilde{x}\|$. Writing $y_t = x_k' + t(x_k - x_k')$ and applying the fundamental theorem of calculus,

$$\Delta e_k = E(x_k) - E(x_k') = \Big(\int_0^1 J_E(y_t)\, \mathrm{d}t\Big)\Delta x_k.$$

Hence

$$\|\Delta e_k\| \le \|\Delta x_k\| \int_0^1 \|J_E(y_t)\|\, \mathrm{d}t \le L_2\|\Delta x_k\| \int_0^1 \|y_t - \tilde{x}\|\, \mathrm{d}t \le L_2\, \max\{\|x_k - \tilde{x}\|, \|x_k' - \tilde{x}\|\}\, \|\Delta x_k\|,$$

using convexity of $t \mapsto \|y_t - \tilde{x}\|$. Finally $\|x_k - \tilde{x}\| \le \|x_k - x_0\| + \|x_0 - \tilde{x}\| \le S + R$, and the same for the primed run, so $\|\Delta e_k\| \le L_2(S+R)\|\Delta x_k\|$. The residual is the stacked vector $\mathcal{R}_k = (-\alpha\Delta e_k, -\Delta e_k)$, whose norm is $\|\mathcal{R}_k\| = \sqrt{1+\alpha^2}\, \|\Delta e_k\|$ exactly; combined with $\|\Delta x_k\| \le \|w_k\|$ this gives $\|\mathcal{R}_k\| \le \sqrt{1+\alpha^2}\, L_2(S+R)\|w_k\|$, which is (7). □

5 Spectral Analysis of the Propagator

The recursion (6) is driven by the non-symmetric matrix $\mathbf{M}$. Because $\mathbf{M}$ is a polynomial in H, it block-diagonalises in the eigenbasis of H into 2×2 blocks, which we analyse one eigen-direction at a time.

Lemma 5.1 (Block reduction)**.** *Let $H = VDV^\top$ with $V = [v_1, \dots, v_d]$ and $D = \mathrm{diag}(\lambda_1, \dots, \lambda_d)$. In the reordered basis $(\Delta x^{(1)}, \Delta d^{(1)}, \dots, \Delta x^{(d)}, \Delta d^{(d)})$, the propagator is block-diagonal, $\mathbf{M} \cong \mathrm{blockdiag}(B_1, \dots, B_d)$, with*

$$B_i = \begin{pmatrix} 1 - \alpha\lambda_i & \alpha\beta \\ -\lambda_i & \beta \end{pmatrix}, \qquad \det B_i = \beta, \qquad \mathrm{tr}\, B_i = 1 - \alpha\lambda_i + \beta. \tag{8}$$

Proof. Conjugating $\mathbf{M}$ by $\mathrm{diag}(V, V)$ replaces H by D in every block of (6), giving $\left(\begin{smallmatrix} I-\alpha D & \alpha\beta I \\ -D & \beta I \end{smallmatrix}\right)$. As all four sub-blocks are diagonal, grouping the i-th coordinate of the x- and d-parts yields the 2×2 block B_i in (8); the determinant and trace are read off directly. $\square$

Lemma 5.2 (Powers of a 2×2 block)**.** *Let B be a 2×2 matrix with eigenvalues μ_1, μ_2 (possibly equal or complex) and spectral radius $\rho = \max\{|\mu_1|, |\mu_2|\}$. For every $n \in \mathbb{N}$,*

$$B^n = a_n B + b_n I, \qquad a_n = \frac{\mu_1^n - \mu_2^n}{\mu_1 - \mu_2}, \qquad b_n = -\mu_1\mu_2 \frac{\mu_1^{n-1} - \mu_2^{n-1}}{\mu_1 - \mu_2},$$

and consequently

$$\|B^n\| \le C(B)\, \rho^n, \qquad C(B) \coloneqq \frac{2\,(\|B\| + \rho)}{|\mu_1 - \mu_2|}. \tag{9}$$

Proof. By the Cayley–Hamilton theorem $B^2 = (\mu_1 + \mu_2)B - \mu_1\mu_2 I$, so $B^n = a_n B + b_n I$ for scalars a_n, b_n. Evaluating on the two eigenvalues, $\mu_i^n = a_n\mu_i + b_n$, and solving the 2×2 linear system gives the stated a_n, b_n. Bounding $|\mu_1^n - \mu_2^n| \le 2\rho^n$ and $|\mu_1\mu_2|\,|\mu_1^{n-1} - \mu_2^{n-1}| \le |\mu_1|\, 2\rho^n$ yields $|a_n| \le 2\rho^n/|\mu_1 - \mu_2|$ and $|b_n| \le |\mu_1|\, 2\rho^n/|\mu_1 - \mu_2|$, hence $\|B^n\| \le |a_n|\|B\| + |b_n| \le \frac{2\rho^n}{|\mu_1 - \mu_2|}(\|B\| + \rho)$. $\square$

5.1 The escape block

The block with the most negative curvature is the engine of the escape. We show it has a real eigenvalue strictly larger than one.

Lemma 5.3 (Escape eigenvalue)**.** *Let $\lambda_{\min} \le -\varepsilon_H$ and let $B_{\min} = B(\lambda_{\min})$ as in (8), with characteristic polynomial $p(\mu) = \mu^2 - t\mu + \beta$, where $t = \mathrm{tr}\, B_{\min} = 1 + \beta + \alpha|\lambda_{\min}|$. Then $B_{\min}$ has real eigenvalues $\mu_- < 1 < \mu_+$ with*

$$\mu_+ \;\ge\; 1 + \alpha|\lambda_{\min}| \;\ge\; 1 + \alpha\varepsilon_H. \tag{10}$$

Proof. Since $\lambda_{\min} < 0$, $t = 1 + \beta + \alpha|\lambda_{\min}| > 1 + \beta$. Evaluating the characteristic polynomial at 1,

$$p(1) = 1 - t + \beta = -\alpha|\lambda_{\min}| < 0,$$

so p has a root on each side of 1, in particular p has two real roots and $\mu_- < 1 < \mu_+$. Writing $a = \alpha|\lambda_{\min}|$ and evaluating at $1 + a$,

$$p(1{+}a) = (1{+}a)^2 - (1{+}\beta{+}a)(1{+}a) + \beta = (1{+}a)\big[(1{+}a) - (1{+}\beta{+}a)\big] + \beta = -\beta(1{+}a) + \beta = -\beta a \le 0.$$

As $p(\mu) \to +\infty$ for $\mu \to \infty$ and $p(1{+}a) \le 0$, the larger root satisfies $\mu_+ \ge 1 + a = 1 + \alpha|\lambda_{\min}| \ge 1 + \alpha\varepsilon_H$. $\square$

Lemma 5.4 (No block grows faster)**.** *Let $\beta > 0$. For every eigenvalue $\lambda \in [\lambda_{\min}, L_1]$ of H, the spectral radius of $B(\lambda)$ satisfies $\rho(\lambda) \le \mu_+$. Consequently, for the propagator,*

$$\|\mathbf{M}^n\| \le C_0\, \mu_+^n, \qquad C_0 \coloneqq \max_i C(B_i). \tag{11}$$

Proof. By (8) the determinant $\det B(\lambda) = \beta$ is constant while $t(\lambda) = 1 - \alpha\lambda + \beta$ is strictly decreasing in λ, hence maximal at $\lambda = \lambda_{\min}$. The eigenvalues are $\mu_\pm(\lambda) = \frac{1}{2}\big(t(\lambda) \pm \sqrt{t(\lambda)^2 - 4\beta}\big)$. If $t(\lambda)^2 < 4\beta$ the eigenvalues are complex conjugates with $|\mu|^2 = \det = \beta$, so $\rho(\lambda) = \sqrt{\beta}$; we have $\rho(\lambda) = \sqrt{\beta} = \sqrt{\mu_+\mu_-} \le \mu_+$. If $t(\lambda)^2 \ge 4\beta$ the larger root $\rho(\lambda) = \frac{1}{2}\big(t(\lambda) + \sqrt{t(\lambda)^2 - 4\beta}\big)$ is increasing in $t(\lambda)$, hence maximal at $\lambda = \lambda_{\min}$, where it equals μ_+. In both cases $\rho(\lambda) \le \mu_+$. Since $\mathbf{M}^n$ is block-diagonal, $\|\mathbf{M}^n\| = \max_i \|B_i^n\| \le \max_i C(B_i)\rho_i^n \le C_0\mu_+^n$ by (9). □

5.2 Growth of the homogeneous part

We decompose the difference state as $w_k = h_k + p_k$, the homogeneous part $h_k = \mathbf{M}^k w_0$ and the residual $p_k = \sum_{\tau=0}^{k-1} \mathbf{M}^{k-1-\tau}\mathcal{R}_\tau$, so that $p_0 = 0$ and $p_{k+1} = \mathbf{M}p_k + \mathcal{R}_k$. With w_0 chosen in the escape plane, the homogeneous part is governed entirely by $B_{\min}$.

Lemma 5.5 (Homogeneous growth)**.** *Let $w_0 = (R_0, 0)$ lie in the two-dimensional escape plane and write, in the eigenbasis of $B_{\min}$, $v_\mu = (\beta - \mu, \lambda_{\min})$, $\zeta = \mu_-/\mu_+ \in [0, 1)$. Fix $\theta \in (0, 1)$ and let k_0 be the least index with $\zeta^{k_0}\|v_-\| \le \theta\|v_+\|$. Then for all $k \ge k_0$,*

$$\|h_k\| \ge c_1\, \mu_+^k R_0, \qquad c_1 \coloneqq (1-\theta)\,\frac{\|v_+\|}{\mu_+ - \mu_-}, \tag{12}$$

and the position block alone satisfies $|\Delta x_k^{\text{hom}}| \ge c_1'\, \mu_+^k R_0$ with $c_1' = (1-\theta)\,|\beta - \mu_+|/(\mu_+ - \mu_-)$. Moreover $\|h_k\| \le c_h\, \mu_+^k R_0$ with $c_h = C(B_{\min})$.

Proof. The eigenvectors of $B_{\min}$ solve $(B_{\min} - \mu I)v = 0$; from the second row, $-\lambda_{\min}v^{(1)} + (\beta - \mu)v^{(2)} = 0$, so $v_\mu = (\beta - \mu, \lambda_{\min})$. Expand $w_0 = \gamma_+ v_+ + \gamma_- v_-$. The second coordinate gives $0 = \gamma_+\lambda_{\min} + \gamma_-\lambda_{\min}$, hence $\gamma_- = -\gamma_+$; the first gives $R_0 = \gamma_+(\beta - \mu_+) + \gamma_-(\beta - \mu_-) = \gamma_+(\mu_- - \mu_+)$, so $|\gamma_+| = R_0/(\mu_+ - \mu_-) \ne 0$. Thus w_0 has a nonzero component along the unstable eigenvector. Therefore

$$h_k = B_{\min}^k w_0 = \gamma_+\mu_+^k v_+ + \gamma_-\mu_-^k v_- = \gamma_+\, \mu_+^k\big(v_+ - \zeta^k v_-\big),$$

and

$$\|h_k\| \ge |\gamma_+|\, \mu_+^k\big(\|v_+\| - \zeta^k\|v_-\|\big) \ge (1-\theta)\,|\gamma_+|\,\|v_+\|\,\mu_+^k = (1-\theta)\frac{\|v_+\|}{\mu_+ - \mu_-}\,\mu_+^k R_0$$

for $k \ge k_0$. Projecting onto the position coordinate replaces $\|v_+\|$ by $|\beta - \mu_+|$, giving the bound on $|\Delta x_k^{\text{hom}}|$. The upper bound is immediate from $\|h_k\| = \|B_{\min}^k w_0\| \le C(B_{\min})\mu_+^k\|w_0\|$ and $\|w_0\| = R_0$. □

6 Geometric Growth of the Gap

It remains to show that the residual p_k does not cancel the growth of h_k. The mechanism is an induction: assuming $\|p_\tau\| \le c\|h_\tau\|$ up to step k, the bound on $\|\mathcal{R}_\tau\|$ keeps the accumulated residual a fraction of h_{k+1}, provided $\sigma_{\text{err}} = \sqrt{1+\alpha^2}\, L_2(S + R)$ is small enough, which is achievable because S and R are at our disposal.

Lemma 6.1 (Geometric growth survives)**.** *Along a momentum window (Definition 4.1), suppose $c \in (0, 1)$ and*

$$\sigma_{\text{err}} \;\le\; \frac{c\, c_1\, \mu_+}{C_0\, c_h\, (1+c)\,(T+1)} \;=:\; \frac{\Lambda}{T+1}. \tag{13}$$

Then $\|p_k\| \le c\,\|h_k\|$ for all $k \le T$, and therefore

$$\|w_T\| \ge (1-c)\, c_1\, \mu_+^T R_0, \qquad \|\Delta x_T\| \ge \big(c_1' - c\, c_h\big)\, \mu_+^T R_0. \tag{14}$$

Proof. We argue by induction on k. The base case $p_0 = 0$ is immediate. Assume $\|p_\tau\| \le c\|h_\tau\|$ for all $\tau \le k$, so that $\|w_\tau\| \le (1+c)\|h_\tau\| \le (1+c)c_h\mu_+^\tau R_0$ by Lemma 5.5. Using $p_{k+1} = \sum_{\tau=0}^{k} \mathbf{M}^{k-\tau}\mathcal{R}_\tau$, the propagator bound (11), the residual bound (7) and the inductive hypothesis,

$$\begin{aligned}\|p_{k+1}\| &\le \sum_{\tau=0}^{k} \|\mathbf{M}^{k-\tau}\|\,\|\mathcal{R}_\tau\| \le \sum_{\tau=0}^{k} C_0\,\mu_+^{k-\tau}\,\sigma_{\text{err}}\,\|w_\tau\| \\ &\le \sum_{\tau=0}^{k} C_0\,\mu_+^{k-\tau}\,\sigma_{\text{err}}\,(1+c)\,c_h\,\mu_+^\tau R_0 = C_0\,c_h\,\sigma_{\text{err}}\,(1+c)\,R_0\,(k+1)\,\mu_+^k.\end{aligned}$$

By (13) and $k+1 \le T+1$, $C_0 c_h \sigma_{\text{err}}(1+c)(k+1) \le c\,c_1\mu_+$, so $\|p_{k+1}\| \le c\,c_1\,\mu_+^{k+1}R_0 \le c\,\|h_{k+1}\|$, the last step by the lower bound (12). This closes the induction. Finally $\|w_T\| \ge \|h_T\| - \|p_T\| \ge (1-c)\|h_T\| \ge (1-c)c_1\mu_+^T R_0$, and projecting onto the position block, $\|\Delta x_T\| \ge |\Delta x_T^{\text{hom}}| - \|p_T\| \ge c_1'\mu_+^T R_0 - c\,c_h\mu_+^T R_0$. □

Remark 6.1. Positivity of the position growth coefficient requires $c < c_1'/c_h$, which is built into the admissible range $c \in (0, \min\{1, c_1'/c_h\})$ of Table 1.

7 Escaping Strict Saddle Points

We now combine the geometric growth of the gap with the confinement of stuck trajectories. The decomposition of the perturbation along and orthogonal to $v_{\min}$ is $\xi = \xi_1 + \xi_\perp$ with $\xi_1 = \langle \xi, v_{\min}\rangle \in \mathbb{R}$ and $\xi_\perp \in v_{\min}^\perp \cong \mathbb{R}^{d-1}$. Define the localized and stuck sets

$$\mathcal{X}_{\text{loc}} = \big\{\xi \in B_0(R) : \|x_k - x_0\| \le S\ \forall\, 0 \le k \le T\big\}, \quad \mathcal{X}_{\text{stuck}} = \big\{\xi \in B_0(R) : f(x_T) - f(x_0) > -F\big\}.$$

By improve-or-localize, a perturbation from which the method fails to decrease the objective by F is localized: $\mathcal{X}_{\text{stuck}} \subseteq \mathcal{X}_{\text{loc}}$. It therefore suffices to bound $\mathbb{P}(\xi \in \mathcal{X}_{\text{loc}})$.

Lemma 7.1 (Threshold separation)**.** *If two stuck trajectories start at x_0 and $x_0' = x_0 + R_0 v_{\min}$, then $\|\Delta x_T\| \le 2S + R_0$. Combined with (14) this forces*

$$R_0 \;\le\; R_0^\star \;:=\; \frac{2S}{(c_1' - c\,c_h)\,\mu_+^T - 1}. \tag{15}$$

Proof. If both runs are stuck, each is confined by Lemma 3.3 to a ball of radius S around its start, so

$$\|\Delta x_T\| \le \|x_T - x_0\| + \|x_0 - x_0'\| + \|x_0' - x_T'\| \le S + R_0 + S = 2S + R_0.$$

On the other hand (14) gives $\|\Delta x_T\| \ge (c_1' - c\,c_h)\mu_+^T R_0$. Hence $(c_1' - c\,c_h)\mu_+^T R_0 \le 2S + R_0$, i.e. $R_0\big[(c_1' - c\,c_h)\mu_+^T - 1\big] \le 2S$, which is (15). □

Lemma 7.2 (Escape with high probability)**.** *Suppose $\|\nabla f(\tilde{x})\| \le \varepsilon_g$ and $\lambda_{\min}(\nabla^2 f(\tilde{x})) \le -\varepsilon_H$, and that a perturbation has not been added in the previous T steps, and that the next T steps form a momentum window (Definition 4.1). If the parameters satisfy*

$$T \;\ge\; \frac{1}{\log \mu_+}\,\log\left[\frac{1}{c_1' - c\,c_h}\left(1 + \frac{2S}{\delta R}\sqrt{\frac{d+1}{2\pi}}\right)\right] \tag{16}$$

and R is chosen so that $\varepsilon_g R + \frac{L_1}{2}R^2 \le \eta F$ with $\eta \le \frac{1}{2}$, then with probability at least $1-\delta$,

$$f(x_T) - f(\tilde{x}) \le -(1-\eta)F \le -\tfrac{1}{2}F. \tag{17}$$

Proof. Volume of the stuck set. By Lemma 7.1, any two stuck perturbations lying on a line parallel to $v_{\min}$ are separated by at most $R_0^\star$, so the one-dimensional Lebesgue measure of $\mathcal{X}_{\mathrm{loc}}$ along each such line is at most $R_0^\star$. By Fubini, splitting $x = (x^{(1)}, x^{(\perp)})$ with $x^{(1)}$ along $v_{\min}$,

$$\mathrm{Vol}(\mathcal{X}_{\mathrm{loc}}) = \int_{B_0^{d-1}(R)} \Big(\int \mathbf{1}\{\mathcal{X}_{\mathrm{loc}}\}\, \mathrm{d}x^{(1)} \Big) \mathrm{d}x^{(\perp)} \le \int_{B_0^{d-1}(R)} R_0^\star \,\mathrm{d}x^{(\perp)} = R_0^\star \,\mathrm{Vol}\big(B_0^{d-1}(R)\big).$$

Using $\mathrm{Vol}(B_0^n(R)) = \pi^{n/2} R^n / \Gamma(\frac{n}{2}+1)$ and $\Gamma(\frac{d}{2}+1)/\Gamma(\frac{d}{2}+\frac{1}{2}) \le \sqrt{(d+1)/2}$,

$$\mathbb{P}(\xi \in \mathcal{X}_{\mathrm{loc}}) = \frac{\mathrm{Vol}(\mathcal{X}_{\mathrm{loc}})}{\mathrm{Vol}(B_0^d(R))} \le \frac{R_0^\star}{R\sqrt{\pi}} \cdot \frac{\Gamma(\frac{d}{2}+1)}{\Gamma(\frac{d}{2}+\frac{1}{2})} \le \frac{R_0^\star}{R}\sqrt{\frac{d+1}{2\pi}}.$$

Requiring the right-hand side to be at most δ is, by the definition (15) of $R_0^\star$, equivalent to $\frac{2S}{\delta R}\sqrt{(d+1)/(2\pi)} + 1 \le (c_1' - c\, c_h)\mu_+^T$, i.e. to (16). Hence $\mathbb{P}(\xi \in \mathcal{X}_{\mathrm{stuck}}) \le \mathbb{P}(\xi \in \mathcal{X}_{\mathrm{loc}}) \le \delta$.

Net decrease. On the complementary event, the trajectory leaves the stuck set, so $f(x_T) - f(x_0) \le -F$. The perturbation itself costs, by L_1-smoothness,

$$f(x_0) - f(\tilde{x}) \le \langle \nabla f(\tilde{x}), \xi \rangle + \frac{L_1}{2}\|\xi\|^2 \le \varepsilon_g R + \frac{L_1}{2} R^2 \le \eta F.$$

Adding the two, $f(x_T) - f(\tilde{x}) \le -F + \eta F = -(1-\eta)F \le -\frac{1}{2}F$ for $\eta \le \frac{1}{2}$. □

8 Iteration Complexity

The escape lemma assumed a constraint system relating T, S, F, R, δ and the growth constants. We record it, show it is solvable, and then assemble the complexity bound and prove Theorem 2.1.

8.1 The constraint system

Collecting the conditions from the previous sections, the operational quantities must satisfy

$$\text{(C1)} \quad S = (1+\nu)\sqrt{\tfrac{TF}{L_1 C(\nu)}} \iff F = \frac{L_1 C(\nu)}{(1+\nu)^2} \cdot \frac{S^2}{T}, \tag{18}$$

$$\text{(C2)} \quad \sigma_{\mathrm{err}} = \sqrt{1+\alpha^2}\, L_2(S+R) \le \frac{\Lambda}{T+1}, \qquad \Lambda = \frac{c\, c_1\, \mu_+}{C_0\, c_h\, (1+c)}, \tag{19}$$

$$\text{(C3)} \quad L_2(S+R) \le \tfrac{1}{2}\varepsilon_H, \tag{20}$$

$$\text{(C4)} \quad \varepsilon_g R + \tfrac{L_1}{2} R^2 \le \eta F, \tag{21}$$

$$\text{(C5)} \quad T = \frac{1}{\log \mu_+} \log\Big[\tfrac{1}{c_1' - c\, c_h}\big(1 + \tfrac{2S}{\delta R}\sqrt{\tfrac{d+1}{2\pi}}\big)\Big]. \tag{22}$$

Here $C(\nu), B_{\min}, \mu_\pm, c_1, c_1', c_h, C_0$ are the derived quantities of Table 1, all functions of the problem data d, L_1, L_2, Δ_f and the free parameters $\varepsilon_g, \varepsilon_H, \beta, \nu, \theta, c, \eta, \delta$ (recall $\alpha = 1/L_1$).

Proposition 8.1 (Solvability)**.** *The system (18)–(22) admits a solution with $R > 0$ and $S > 0$. Concretely, setting $\Psi \coloneqq S + R = \Lambda/(\sqrt{1+\alpha^2}\, L_2(T+1))$ and*

$$R = \min\left\{ \frac{\eta F}{2\varepsilon_g},\ \sqrt{\frac{\eta F}{L_1}},\ \frac{\Lambda}{2\sqrt{1+\alpha^2}\, L_2(T+1)} \right\}, \qquad S = \Psi - R \ge \frac{\Lambda}{2\sqrt{1+\alpha^2}\, L_2(T+1)} > 0,$$

verifies (C2) with equality and (C4) by splitting each summand below $\frac{1}{2}\eta F$; (C3) follows from (C2) once $T \ge 2\Lambda/\varepsilon_H - 1$, which holds because $\Lambda = \Theta(1)$ and $T = \Theta(L_1\phi/\varepsilon_H) \gg 1/\varepsilon_H$; and (C1) then determines F.

Proof. Set (C2) to equality, giving $\Psi = \Lambda/(\sqrt{1+\alpha^2}\,L_2(T+1))$. The first two terms in the definition of R enforce (C4): since $\frac{L_1}{2}R^2 \le \frac{1}{2}\eta F$ when $R \le \sqrt{\eta F/L_1}$ and $\varepsilon_g R \le \frac{1}{2}\eta F$ when $R \le \eta F/(2\varepsilon_g)$, their sum is at most ηF. The third term enforces $R \le \frac{1}{2}\Psi$, when $S = \Psi - R \ge \frac{1}{2}\Psi > 0$; any split fraction in $(0,1)$ would serve, the requirement being only $R > 0$ (a genuine perturbation) and $S > 0$ (a non-vacuous localization). With (C2) at equality, $\sigma_{\mathrm{err}} = \sqrt{1+\alpha^2}\,L_2\Psi = \Lambda/(T+1)$, so $L_2(S+R) = \Lambda/(\sqrt{1+\alpha^2}\,(T+1)) \le \Lambda/(T+1)$ and (C3) holds once $\Lambda/(T+1) \le \varepsilon_H/2$, i.e. $T+1 \ge 2\Lambda/\varepsilon_H$; as $\Lambda = \Theta(1)$ and $T = \Theta(L_1\phi/\varepsilon_H)$ with $\phi \gg 1$, this holds. Finally (C1) defines F from the chosen S, T. □

Proposition 8.2 (Scales)**.** *With the choice of Proposition 8.1 and* $\phi := \log\big[(c_1' - c\,c_h)^{-1}\big] + \log\big(1 + \frac{2S}{\delta R}\sqrt{(d+1)/(2\pi)}\big) = \Theta\big(\log\frac{d\,L_1L_2\Delta_f}{\varepsilon_H\,\delta}\big)$*,*

$$T = \Theta\Big(\frac{L_1\phi}{\varepsilon_H}\Big), \quad S = \Theta\Big(\frac{\varepsilon_H}{L_1L_2\phi}\Big), \quad F = \Theta\Big(\frac{\varepsilon_H^3}{L_1^2L_2^2\phi^3}\Big).$$

Proof. By Lemma 5.3, $\mu_+ \ge 1 + \alpha\varepsilon_H$ with $\alpha = 1/L_1$, and $\log(1+x) \ge x/2$ on $[0,1]$ gives $\log\mu_+ \ge \frac{1}{2}\alpha\varepsilon_H$, so by (22) $T \le 2\phi/(\alpha\varepsilon_H) = \Theta(L_1\phi/\varepsilon_H)$. Then $S = \Theta(\Lambda/(\sqrt{1+\alpha^2}\,L_2T)) = \Theta(\varepsilon_H/(L_1L_2\phi))$, since Λ and $\sqrt{1+\alpha^2}$ are constants in the tolerances, and (18) gives $F = \Theta(L_1S^2/T) = \Theta\big(L_1 \cdot \frac{\varepsilon_H^2}{L_1^2L_2^2\phi^2} \cdot \frac{\varepsilon_H}{L_1\phi}\big) = \Theta(\varepsilon_H^3/(L_1^2L_2^2\phi^3))$. □

8.2 Counting the iterations

Proof of Theorem 2.1. We split a run into large-gradient steps and escape episodes.

Large-gradient phase. While $\|g_k\| \ge \varepsilon_g$, no perturbation is triggered and Lemma 3.2 gives $f(x_k) - f(x_{k+1}) \ge \frac{C(\nu)}{L_1}\|g_k\|^2 > \frac{C(\nu)}{L_1}\varepsilon_g^2$. Since the total decrease available is Δ_f, the number of such steps is at most

$$N_{\mathrm{desc}} \le \frac{L_1\Delta_f}{C(\nu)\,\varepsilon_g^2} = O(\varepsilon_g^{-2}). \tag{23}$$

Escape phase. Each perturbation episode that does not certify a 2OSP produces, by Lemma 7.2, a net decrease of at least $(1-\eta)F$. Here Lemma 7.2 is applied to the momentum window of each episode (Definition 4.1). Hence the number of episodes is at most $\bar{K} \le \Delta_f/((1-\eta)F) + 1$, and by the scale of F in Proposition 8.2,

$$\bar{K} = \Theta\Big(\frac{L_1^2L_2^2\Delta_f\,\phi^3}{\varepsilon_H^3}\Big) = \widetilde{O}(\varepsilon_H^{-3}).$$

Each episode costs T iterations, so

$$N_{\mathrm{esc}} \le T\,\bar{K} = \Theta\Big(\frac{L_1\phi}{\varepsilon_H}\Big)\cdot\Theta\Big(\frac{L_1^2L_2^2\Delta_f\phi^3}{\varepsilon_H^3}\Big) = \Theta\Big(\frac{L_1^3L_2^2\Delta_f\,\phi^4}{\varepsilon_H^4}\Big) = \widetilde{O}(\varepsilon_H^{-4}). \tag{24}$$

Union bound. Index the episodes by $j = 1,\dots,\bar{K}$; episode j takes a snapshot $\tilde{x}^{(j)}$ with $\|\nabla f(\tilde{x}^{(j)})\| \le \varepsilon_g$ and runs a window of length T. Define the bad event $\mathcal{E}_j = \{\lambda_{\min}(\nabla^2 f(\tilde{x}^{(j)})) \le -\varepsilon_H$ and the window fails to decrease by $F\}$; that is, $\tilde{x}^{(j)}$ was a strict saddle from which the method did not escape. By Lemma 7.2, $\mathbb{P}(\mathcal{E}_j) \le \delta$ for every j. Fixing a total budget $\delta_{\mathrm{tot}} \in (0,1)$ and setting $\delta = \delta_{\mathrm{tot}}/\bar{K}$, the union bound gives

$$\mathbb{P}\Big(\textstyle\bigcup_{j=1}^{\bar{K}}\mathcal{E}_j\Big) \le \textstyle\sum_{j=1}^{\bar{K}}\mathbb{P}(\mathcal{E}_j) \le \bar{K}\,\delta = \delta_{\mathrm{tot}}.$$

With probability at least $1-\delta_{\mathrm{tot}}$, then, no saddle episode fails. This choice only affects the logarithmic factor: since $\log(1/\delta) = \log(\bar{K}/\delta_{\mathrm{tot}}) = \log\bar{K} + \log(1/\delta_{\mathrm{tot}})$ and $\log\bar{K} = \Theta(\log(1/\varepsilon_H)) = O(\phi)$, it is absorbed into ϕ, preserving the scales of Proposition 8.2.

Summing (23) and (24) yields $N = O(\varepsilon_g^{-2}) + \widetilde{O}(\varepsilon_H^{-4})$, completing the proof. □

9 Conclusion

We have shown that a perturbed, restarted constant-momentum method—equivalently, a safeguarded heavy-ball iteration—escapes strict saddle points and reaches an $(\varepsilon_g, \varepsilon_H)$-second-order stationary point in $O(\varepsilon_g^{-2}) + \widetilde{O}(\varepsilon_H^{-4})$ iterations, matching the rate of perturbed gradient descent. The analysis transfers the perturbation-and-coupling framework of [3, 4] to an iteration whose linearised propagator is non-symmetric, via an explicit spectral study of the two-trajectory difference.

The momentum coefficient β was held constant throughout, and this is the main limitation of the present result. The natural next step, and our motivating goal, is the genuine nonlinear conjugate gradient method, in which $\beta_k = \|g_k\|^2/\|g_{k-1}\|^2$ (Fletcher–Reeves [2]) varies with the iterates and is singular at stationary points, where the denominator $\|g_{k-1}\|^2$ vanishes. That singularity is precisely what obstructs a direct extension of the propagator analysis—the block $B_{\min}$ would itself depend on k through β_k—and treating it appears to require new ideas. Another interesting future direction would consist in adapting the argument to a line-search regime, where Wolfe conditions should supply the angle and decrease guarantees that the fixed step size and restart test furnish here.

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Symbol	Meaning	Definition / constraint
ν	restart slack	$0 < \nu < \sqrt{5} - 2$
$C(\nu)$	decrease constant	$(1-\nu) - \frac{1}{2}(1+\nu)^2 = \frac{1}{2}(1 - 4\nu - \nu^2)$
β	momentum (constant)	$\beta \in (0, 1]$
θ	warm-up margin	$0 < \theta < 1$
c	residual fraction	$0 < c < \min\{1, c_1'/c_h\}$
η	perturbation-cost fraction	$0 < \eta \leq \frac{1}{2}$
δ	per-episode failure prob.	$\delta = \delta_{\mathrm{tot}}/\bar{K}$
ε_g	gradient tolerance	$\varepsilon_g > 0$
ε_H	curvature tolerance	$0 < \varepsilon_H \leq L_1$
α	step size	$1/L_1$
H, λ_i	$\nabla^2 f(\tilde{x})$ and its eigenvalues	$\lambda_{\min} \leq -\varepsilon_H$
B_i	$\left(\begin{smallmatrix} 1-\alpha\lambda_i & \alpha\beta \\ -\lambda_i & \beta \end{smallmatrix}\right)$	$\det B_i = \beta$
$\mu_\pm$	eigenvalues of $B_{\min}$	$\mu_+ \geq 1 + \alpha\varepsilon_H > 1 > \mu_-$
ρ_i	spectral radius of B_i	$\max\{\lvert\mu_1\rvert, \lvert\mu_2\rvert\} \leq \mu_+$
$C(B)$	power constant	$2(\lVert B\rVert + \rho)/\lvert\mu_1 - \mu_2\rvert$
c_h, C_0	block constants	$c_h = C(B_{\min}) \leq C_0 = \max_i C(B_i)$
c_1	energy growth	$(1-\theta)\lVert v_+\rVert/(\mu_+ - \mu_-)$
c_1'	position growth	$(1-\theta)\lvert\beta - \mu_+\rvert/(\mu_+ - \mu_-)$
σ_{err}	residual coefficient	$\sqrt{1+\alpha^2}\, L_2(S+R)$
S	localization radius	$(1+\nu)\sqrt{TF/(L_1 C(\nu))}$
F	escape decrease	$L_1 C(\nu)\, S^2/((1+\nu)^2 T)$
R	perturbation radius	Proposition 8.1
$R_0, R_0^\star$	separation, threshold	$R_0^\star = 2S/((c_1' - c\, c_h)\mu_+^T - 1)$
T	escape window	(22); $\Theta(L_1\phi/\varepsilon_H)$
ϕ	log factor	$\Theta(\log(dL_1L_2\Delta_f/(\varepsilon_H\delta)))$
$\bar{K}$	episode count	$\widetilde{O}(\varepsilon_H^{-3})$

Table 1: Free parameters, problem data and derived quantities.